AArch64 Memory & Cache System — Technical Reference Workbook 1
Chapter 3 — Cache Maintenance Instructions (DC/IC): Architectural Semantics and Correct Usage (Based on ARM64 CortexA72) 1
3.1 Scope 1
3.2 Visibility Domains Relevant to Cache Maintenance 1
3.3 Data Cache Maintenance Instructions (DC) 1
3.3.1 DC CVAU — Clean to Point of Unification (PoU) 2
3.3.2 DC CVAC — Clean to Point of Coherence (PoC) 2
3.3.3 DC CIVAC — Clean + Invalidate to PoC 2
3.3.4 DC IVAC — Invalidate to PoC 2
3.3.5 DC ZVA — Zero Cache Line 2
3.4 Instruction Cache Maintenance Instructions (IC) 3
3.4.1 IC IVAU — Invalidate Instruction Cache Line to PoU 3
3.4.2 IC IALLU / IALLUIS 3
3.5 Required Barrier Instructions 3
3.5.1 DSB — Data Synchronization Barrier 3
3.5.2 ISB — Instruction Synchronization Barrier 3
3.6 Correct Usage Patterns 4
3.6.1 Instruction Modification (SelfModifying Code, JIT, Bootloader Patching) 4
3.6.2 CPU → DMA (Device Reads CPU Data) 4
3.6.3 DMA → CPU (Device Writes Data) 4
3.6.4 SMP BringUp (CPU0 Prepares Code for CPU1) 4
3.6.5 MMIO / Device Registers 5
3.7 Microarchitectural Notes (CortexA72) 5
3.8 Exercises (Technical Reference) 5
Exercise 3.1 — Instruction Patching 5
Exercise 3.2 — DMA Read Path 5
Exercise 3.3 — DMA Write Path 6
Exercise 3.4 — Barrier Semantics 6

Exercise 3.5 — MMIO..... 6

AArch64 Memory & Cache System — Technical Reference Workbook

Chapter 3 — Cache Maintenance Instructions (DC/IC): Architectural Semantics and Correct Usage (Based on ARM64 Cortex-A72)

3.1 Scope

This chapter defines the architectural semantics of AArch64 cache maintenance instructions, focusing on Cortex-A72 behavior. It describes:

- Data cache maintenance operations (DC)
- Instruction cache maintenance operations (IC)
- Required ordering constraints
- Visibility domain transitions (L1D → L2 → PoC)
- Correct usage patterns for instruction patching, DMA, and SMP bring-up

All semantics follow ARMv8-A architecture and Cortex-A72 microarchitectural behavior.

3.2 Visibility Domains Relevant to Cache Maintenance

Cache maintenance instructions operate on transitions between visibility domains:

Domain	Description	Relevant Instructions
L1D	Core-local data cache	DC CVAU, DC CVAC, DC CIVAC
L1I	Core-local instruction cache	IC IVAU
L2 (PoU)	Cluster-shared unified domain	DC CVAU, IC IVAU
PoC	System-wide coherent domain	DC CVAC, DC CIVAC

Cache maintenance instructions do **not** operate on MMIO or Device-nGnRnE memory.

3.3 Data Cache Maintenance Instructions (DC)

3.3.1 DC CVAU — Clean to Point of Unification (PoU)

Purpose: Push dirty data from L1D to L2 (PoU).

Architectural effect:

- Ensures instruction refetch sees updated bytes
- Required before IC IVAU when modifying instructions
- Does **not** guarantee visibility to DMA

Cortex-A72 behavior: Triggers L2 snoop of L1D via duplicate directory.

3.3.2 DC CVAC — Clean to Point of Coherence (PoC)

Purpose: Push dirty data from L1D/L2 to PoC (DRAM/interconnect).

Architectural effect:

- Required before DMA reads CPU-generated data
- Ensures external agents observe updated memory

3.3.3 DC CIVAC — Clean + Invalidate to PoC

Purpose: Write back dirty data to PoC and invalidate local copies.

Architectural effect:

- Required after DMA writes to memory
- Ensures CPU refetches fresh data from PoC
- Prevents stale L1D/L2 lines from being used

3.3.4 DC IVAC — Invalidate to PoC

Purpose: Invalidate local cache line without writeback.

Architectural effect: Used when DMA writes to memory and the line is known to be clean.

3.3.5 DC ZVA — Zero Cache Line

Purpose: Efficiently zero a cache line.

Architectural effect: Allocates a zeroed line in L1D; may allocate in L2 depending on policy.

3.4 Instruction Cache Maintenance Instructions (IC)

3.4.1 IC IVAU — Invalidate Instruction Cache Line to PoU

Purpose: Invalidate stale instruction bytes in L1I.

Architectural effect:

- Forces next instruction fetch to consult L2 (PoU)
- Required after modifying instructions in memory
- Must be paired with DC CVAU + DSB + ISB

Cortex-A72 behavior: Does not affect front-end pipeline; ISB required.

3.4.2 IC IALLU / IALLUIS

Purpose: Invalidate entire instruction cache.

Architectural effect: Used during early boot or context switches.

3.5 Required Barrier Instructions

3.5.1 DSB — Data Synchronization Barrier

Purpose: Ensure completion of all cache maintenance operations.

Architectural effect:

- Drains FEQ (Fill/Evict Queue)
- Completes snoops and writebacks
- Enforces ordering
- Required after DC CVAU, DC CVAC, DC CIVAC

3.5.2 ISB — Instruction Synchronization Barrier

Purpose: Flush front-end pipeline and force instruction refetch.

Architectural effect: Required after IC IVAU to ensure updated instructions are executed.

3.6 Correct Usage Patterns

3.6.1 Instruction Modification (Self-Modifying Code, JIT, Bootloader Patching)

Required sequence:

```
DC   CVAU, xN
IC   IVAU, xN
DSB  ISH
ISB
```

Ensures updated instructions propagate from L1D → L2 → L1I → front-end.

3.6.2 CPU → DMA (Device Reads CPU Data)

Required sequence:

```
DC   CVAC, xN
DSB  ISH
```

Ensures DMA sees updated data at PoC.

3.6.3 DMA → CPU (Device Writes Data)

Required sequence:

```
DC   CIVAC, xN
DSB  ISH
```

Ensures CPU refetches fresh data from PoC.

3.6.4 SMP Bring-Up (CPU0 Prepares Code for CPU1)

Required sequence on CPU0:

```
DC   CVAU, xN
IC   IVAU, xN
DSB  ISH
ISB
```

Ensures secondary cores execute updated bootstrap code.

3.6.5 MMIO / Device Registers

- Mark region Device-nGnRnE
- Do **not** use DC/IC
- Use DSB/ISB for ordering only

3.7 Microarchitectural Notes (Cortex-A72)

Although hidden behind architectural barriers, Cortex-A72 implements:

- Duplicate directory snoops for DC CVAU/CVAC/CIVAC
- FEQ pipelining for cache maintenance
- Non-inclusive, non-exclusive L2 behavior
- No L1I snooping (instruction refetch required)
- Register slices for large L2 configurations (timing only)

These details do not change architectural semantics.

3.8 Exercises (Technical Reference)

Exercise 3.1 — Instruction Patching

Explain why DC CVAU must precede IC IVAU when modifying instructions.

Exercise 3.2 — DMA Read Path

Describe the architectural reason DC CVAC is required before DMA reads CPU-generated data.

Exercise 3.3 — DMA Write Path

Explain why DC CIVAC is required after DMA writes to memory.

Exercise 3.4 — Barrier Semantics

Describe the difference between DSB and ISB in the context of instruction modification.

Exercise 3.5 — MMIO

Explain why DC/IC operations must not be used on Device-nGnRnE regions.